

Circular Dependency



```
class OrderService {  
  
    @Autowired  
    PaymentService psi;  
  
}
```



Spring IOC container

1. Create OrderService empty obj.
2. Create PaymentService empty obj
3. Inject paymentService to OrderService
4. Inject orderService to PaymentService.

A - OrderService

B - PaymentService

IOC Container

1. Create A.

2. Inject its dependency

3. Create B

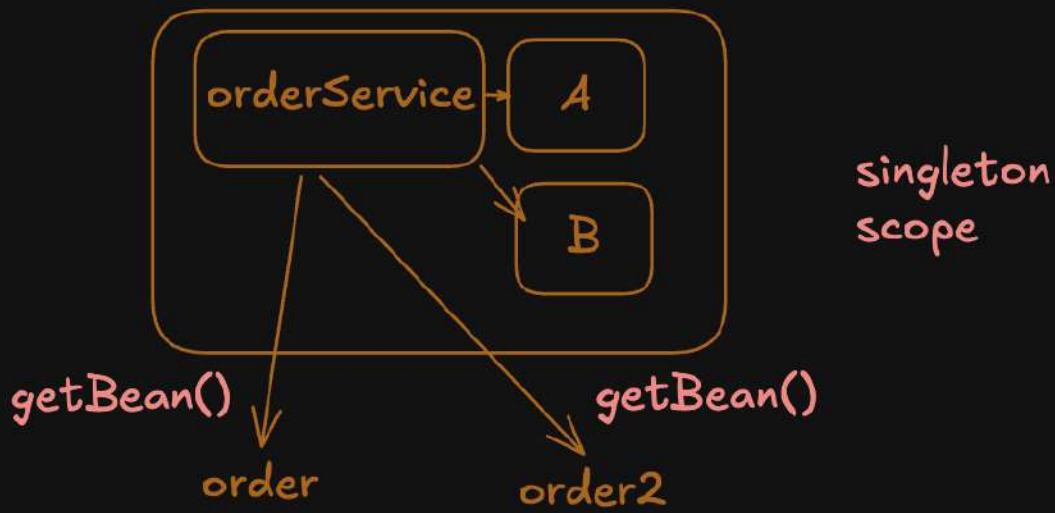
4. Inject dependency



IOC Container

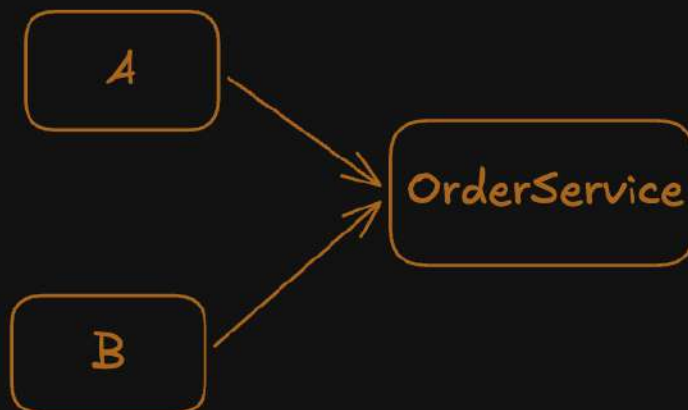
`spring.main.allow-circular-reference : false`

Bean Scopes

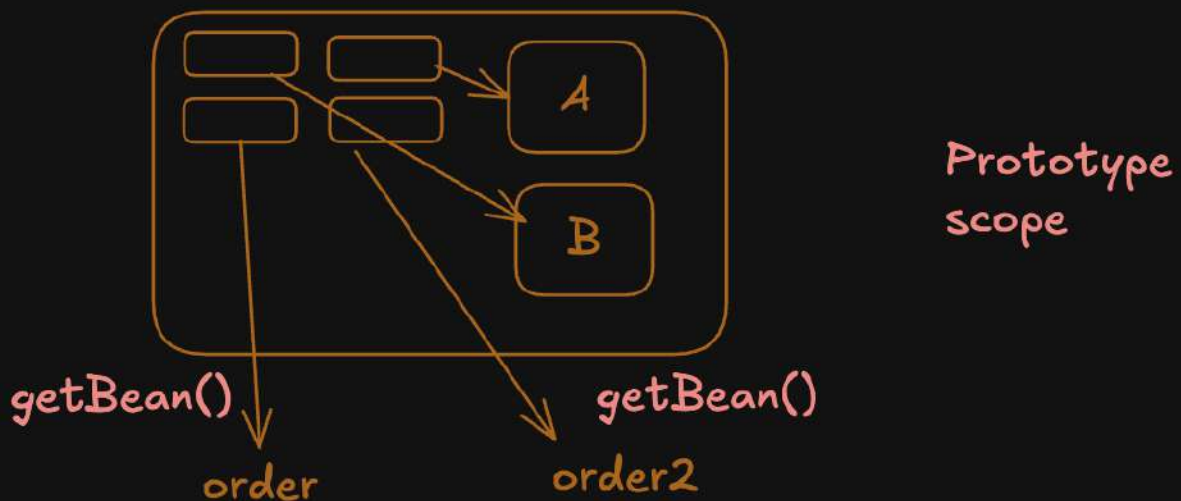


Bean Scopes :

1. Singleton
2. Prototype



Bean Scopes



Singleton --> Eager initialization
Prototype --> Lazy initialization

Singleton : stateless
Prototype : Stateful

Other scopes :

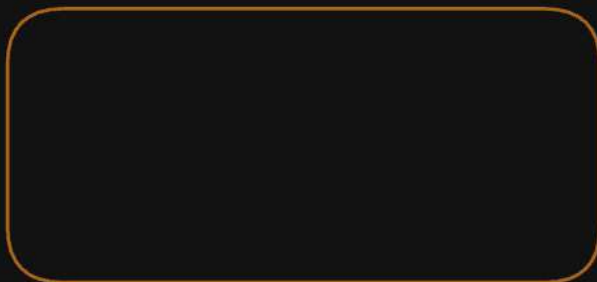
- Request
- Session
- Application

Beans Initialization :

- > Eager
- > Lazy



Eager Initialization

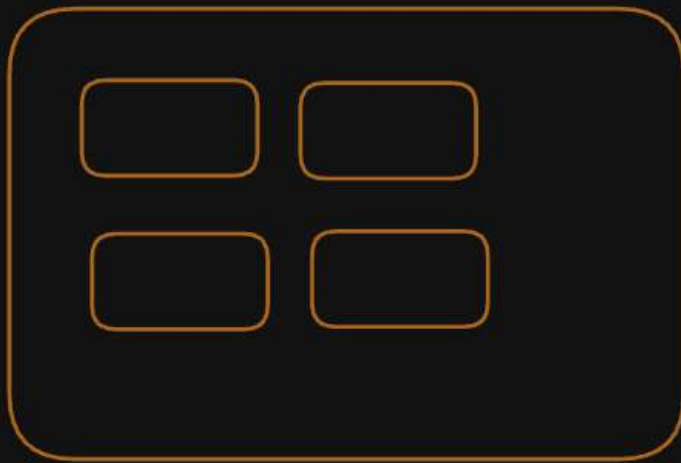


Lazy initialization

Singleton -> Eager, Lazy (@Lazy)

Prototype -> Lazy

OrderService --> Proxy

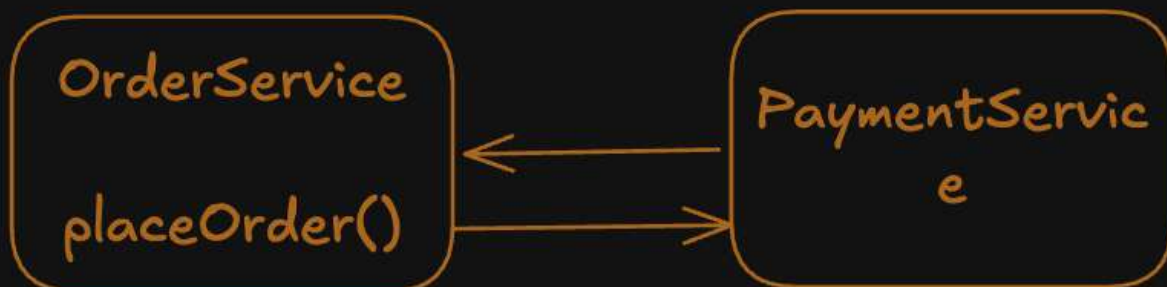


Fail fast

`spring.main.lazy-initialization = true`

`@Lazy("false")`

Circular dependency --> `@Lazy`



proxy